

Questions with Answer Keys

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Q1 - Geometry

The locus of z in complex plane satisfying the condition $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{4}$, is

- (1) a circle with centre (0,0) and radius 2
- (2) a straight line
- (3) a circle with centre (0,0) and radius 3
- (4) None of the above

Q2 - Geometry

In the region $|z + 1 - i| \leq 1$, which of the following complex number has least positive argument?

- (1) i
- (2) $1 + i$
- (3) $-i$
- (4) $-1 + i$

Q3 - Geometry

The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices

- (1) of area zero
- (2) right angled isosceles triangle
- (3) equilateral
- (4) obtuse angled isosceles

Q4 - Geometry

If z and w are two non-zero complex numbers such that $|zw| = 1$ and $\arg(z) - \arg(w) = \frac{\pi}{2}$, then $\bar{z}w$ is equal to

- (1) 1
- (2) -1
- (3) i

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(4) $-i$

Q5 - Geometry

Let z, w be two complex numbers such that $\bar{z} + i\bar{w} = 0$ and $\arg(zw) = \pi$. Then, $\arg(z)$ equals

(1) $\frac{\pi}{4}$

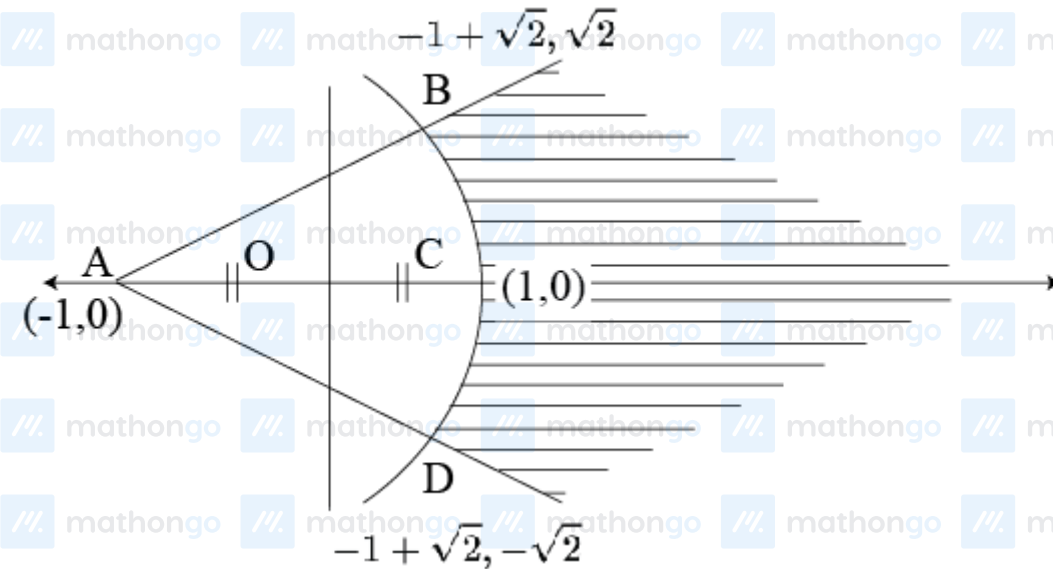
(2) $\frac{\pi}{2}$

(3) $\frac{3\pi}{4}$

(4) $\frac{5\pi}{4}$

Q6 - Geometry

The locus of z , which lies in shaded region (excluding the boundaries) is best, represented by



(1) $z : |z + 1| > 2$ and $|\arg(z + 1)| < \frac{\pi}{4}$

(2) $z : |z - 1| > 2$ and $|\arg(z - 1)| < \frac{\pi}{4}$

(3) $z : |z - 1| > 2$ and $|\arg(z + 1)| < \frac{\pi}{2}$

(4) $z : |z - 1| < 2$ and $|\arg(z + 1)| < \frac{\pi}{2}$

Q7 - Geometry

If $z_1 = 2 + 3i$, $z_2 = 3 - 2i$ and $z_3 = -1 - 2\sqrt{3}i$, then which of the following is true?

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$$(1) \arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$$

$$(2) \arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_2}{z_1}\right)$$

$$(3) \arg\left(\frac{z_3}{z_2}\right) = 2 \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$$

$$(4) \arg\left(\frac{z_3}{z_2}\right) = \frac{1}{2} \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$$

Q8 - Geometry

The number of complex numbers z such that $|z| = 1$ and $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$ is

(1) 4

(2) 6

(3) 8

(4) more than 8

Q9 - Nth root of unity

If $\cos \theta + i \sin \theta$ is a root of equation $x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n = 0$

Then, the value of $\sum_{r=1}^n a_r \cos r\theta$ is

(1) 1

(2) 0

(3) None of these

(4) -1

Q10 - Nth root of unity

If z satisfies $z + \frac{1}{z} = 2 \cos 13^\circ$, then find an angle B , so that $0 < B < \frac{\pi}{2}$ and $z^2 + \frac{1}{z^2} = 2 \cos B$

(1) 23°

(2) 24°

(3) 25°

(4) 26°

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Q11 - Nth root of unity

If $\alpha = e^{i(2\pi/n)}$, then $(11 - \alpha)(11 - \alpha^2) \cdots (11 - \alpha^{n-1})$ is equal to

(1) 11^{n-1}

(2) $\frac{11^n - 1}{10}$

(3) $\frac{11^{n-1} - 1}{10}$

(4) $\frac{11^{n-1} - 1}{11}$

Q12 - Nth root of unity

If z is a complex number such that $z + \frac{1}{z} = 2 \cos 3^\circ$, then the value of $z^{2000} + \frac{1}{z^{2000}} + 1$ is

(1) 0

(2) -1

(3) $\sqrt{3} + 1$

(4) $1 - \sqrt{3}$

Q13 - Nth root of unity

If the expression $x^{2m} + x^m + 1$ is divisible by $x^2 + x + 1$, then

(1) m is an odd integer

(2) m is divisible by 3

(3) m is not divisible by 3

(4) None of the above

Q14 - Nth root of unity

The value of $(x - 1) \left(x + \frac{1}{2} - \frac{i\sqrt{3}}{2} \right) \left(x + \frac{1}{2} + \frac{i\sqrt{3}}{2} \right)$ is

(1) $x^3 + x^2 + x - 1$

(2) $x^3 - 1$

(3) $x^3 + 1$

(4) $x^3 - x^2 + x + 1$

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Q15 - Nth root of unity

If $z^2 + z + 1 = 0$, where z is a complex number then

$\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + \dots + \left(z^6 + \frac{1}{z^6}\right)^2$ is equal to

(1) 54

(2) 6

(3) 12

(4) 18

Q16 - Nth root of unity

If z and w are two complex numbers simultaneously satisfying the equations, $z^3 + w^5 = 0$ and $z^2 \cdot \bar{w}^4 = 1$, then

(1) z and w both are purely real

(2) z is purely real and w is purely imaginary

(3) w is purely real and z is purely imaginary

(4) z and w both are imaginary

Q17 - Nth root of unity

Let $z = \cos \theta + i \sin \theta$, then the value of $\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1})$ at $\theta = 2^\circ$ is

(1) $\frac{1}{\sin 2}$

(2) $\frac{1}{3 \sin 2^\circ}$

(3) $\frac{1}{2 \sin 2^\circ}$

(4) $\frac{1}{4 \sin 2^\circ}$

Q18 - Rotation of complex numbers

The triangle formed by z , iz and $i^2 z$, is

(1) right angled

(2) equilateral

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(3) isosceles

(4) right angled isosceles

Q19 - Rotation of complex numbers

The centre of a square is at origin and one of the vertex is $(1 - i)$. The extremities of diagonal not passing through its vertex are

(1) $1 + i, -1 - i$ (2) $-1 + i, -1 - i$ (3) $1 + i, -1 + i$

(4) None of these

Q20 - Rotation of complex numbers

The centre of a regular polygon of n sides is located at a point $z = 0$ and one of its vertex z_1 is known. If z_2 is the vertex adjacent to z_1 , then z_2 is equal to

(1) $z_1 \left(\cos \frac{2\pi}{n} \pm i \sin \frac{2\pi}{n} \right)$ (2) $z_1 \left(\cos \frac{\pi}{n} \pm i \sin \frac{\pi}{n} \right)$ (3) $z_1 \left(\cos \frac{\pi}{2n} \pm i \sin \frac{\pi}{2n} \right)$

(4) None of these

Q21 - Rotation of complex numbers

Let z_1, z_2 and z_3 be three vertices of an equilateral triangle circumscribing the circle $|z| = \frac{1}{2}$. If $z_1 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$

and z_1, z_2, z_3 are in anti-clockwise sense, then z_2 is equal to

(1) $1 + i\sqrt{3}$ (2) $1 - i\sqrt{3}$

(3) 1

(4) -1

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Q22 - Rotation of complex numbers

If z_1, z_2 and z_3 are vertices of an isosceles triangle and right angled at z_2 , then

(1) $z_1^2 + z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$

(2) $z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$

(3) $2z_1^2 + z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$

(4) $z_1^2 + z_2^2 + 2z_3^2 = 2z_2(z_1 + z_3)$

Q23 - Rotation of complex numbers

A man walks a distance 3 units from the origin towards the North-East ($N45^\circ E$) direction. From there, he walks a distance of 4 units towards the North-West ($N45^\circ W$) direction to reach a point P . Then, the position P in the argand plane is

(1) $3e^{i\pi/4} + 4i$

(2) $(3 - 4i)e^{i\pi/4}$

(3) $(4 + 3i)e^{i\pi/4}$

(4) $(3 + 4i)e^{i\pi/4}$

Q24 - Rotation of complex numbers

If z_1 and $\bar{z}_1$ represents adjacent vertices of a regular polygon of n sides with centre at the origin and

$\frac{\operatorname{Im}(z_1)}{\operatorname{Re}(z_1)} = \sqrt{2} - 1$, then the value of n is

(1) 8

(2) 12

(3) 16

(4) 24

Q25 - Rotation of complex numbers

If w_1 and w_2 are the complex slope of two lines on the argand plane, then the straight lines are perpendicular, if

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(1) $w_1 w_2 = -1$	mathongo	mathongo	mathongo	mathongo	mathongo	mathongo
(2) $w_1 + w_2 = 0$	mathongo	mathongo	mathongo	mathongo	mathongo	mathongo
(3) $w_1 = w_2$	mathongo	mathongo	mathongo	mathongo	mathongo	mathongo
(4) None of these	mathongo	mathongo	mathongo	mathongo	mathongo	mathongo
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